

Introduction to SketchEngine and Web as Corpus research



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In this video you will be introduced to

- SketchEngine web tool for corpus linguistics
- WebBootCat tool in SkE to collect a corpus from the Web
- how to use SkE to collect and analyse text data from the Web
- Leeds Internet Corpora with seed-words for a large corpus
- ACL SIGWAC: Association for Computational Linguistics Special Interest Group on Web as Corpus.

Texts you should read for part 1.3:

- A Kilgariff et al. 2014. The SketchEngine ten years on.
- Websites – see Minerva learning resources

Word sketch – summary of a word's collocational behaviour

Concordance – examples of a word, lemma, phrase in context

Distributional Thesaurus – words with similar meaning / context

Parallel corpus – bilingual corpus, for translation examples

WebBootCat – create specialized corpus from web-pages

Terminology – key words and multi-word terms in a specialism

And more ...

Word sketch – summary of a word's collocational behaviour



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catch

/verb/ Alternative PoS: noun (99054)
enTenTen12 freq = 1111175 (101.1 per million)

sketch	211,096	0.4	sketch	212,711	0.3	sketch	21,712	0.0	sketch	211,411	0.4	sketch	211,479	0.3
glimpse	17,529	9.34	angler	514	0.3	prosecute	201	0.11	up	215,751	0.74	myself	4,170	0.02
fish	16,718	8.70	eye	11,526	6.10	pursue	620	0.33	on	7,200	7.21	him	17,361	0.50
attention	17,300	9.29	fisherman	860	0.47	overhaul	51	0.03	out	2,814	1.40	her	6,421	0.22
breath	11,476	6.10	surprise	1,121	0.62	convict	200	0.10	along	170	1.44	himself	279	0.01
eye	46,715	7.9	touchdown	620	0.33	boat	127	0.06	away	161	1.29	himself	1,520	0.05
pen	7,151	7.44	Anyone	860	0.47	farm	51	0.03	through	50	1.44	we	16,750	0.51
sight	8,200	7.54	police	1,415	0.8	jail	61	0.03	open	20	1.10	them	16,261	0.46
five	11,426	7.39	'I	174	0.09	ingrains	51	0.03				yourself	1,300	0.04
trout	1,600	7.10	breath	330	0.17	tangle	67	0.03				sometimes	230	0.00
but	5,175	6.49	fish	1,001	0.50	same	20	0.76				so	1,351	0.20
ham	1,601	0.40	TO	171	0.09	three	1,200	0.71				him	50	0.07
ball	5,172	0.81	fisher	181	0.15	trap	20	0.04				it	11,000	0.12

sketch	85,204	0.4	sketch	21,511	0.1	sketch	21,712	0.4	sketch	1,402	0.1	sketch	1,402	0.1
crossfire	1,211	0.43	videotape	181	0.09	surprise	1,281	0.79	part	611	0.77	red-handed	201	0.41
trap	1,203	7.76	tape	1,416	0.17	fisherman	171	0.54	hooker	20	0.01	unprepared	270	7.76
middle	1,116	0.49	CCTV	101	0.74	paperwork	10	0.04	centralised	10	0.76	flattened	20	7.01
headlight	511	0.49	wildfire	101	0.71	angler	100	0.47	hit	17	0.40	aware	270	0.09
shutley	200	0.01	five	1,300	0.7	it	11	0.11	prostitute	61	0.1	red-handed	11	0.06
cross-fire	171	0.76	ig	20	0.07	hooker	17	0.04	crime	2	0.11	one-handed	11	0.07
more	211	0.51	telescope	11	0.04	victim	10	0.01	alcohol	601	0.01	off balance	11	0.09
throat	1,217	0.24	misses	61	0.01	fisher	81	0.74	baggage	7	0.01	unprepared	10	0.1
downpour	120	0.10	cameras	1,111	0.51	avalanche	61	0.42	trifles	1	0.04	single-handed	1	0.14
traps	111	0.0	ball	170	0.9	father	11	0.13	eyes	9	0.16	innocently	9	0.00
crosshairs	111	0.0	here	101	0.87	catcher	20	0.04	provisions	1	0.11	harm-handed	1	0.04
avalanche	100	0.0	videotape	17	0.79	petition	11	0.01	staff	9	0.11	sharp-handed	9	0.79

Word sketch – summary of a word's collocational behaviour (Arabic)



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أخضر arTenTen12 [sample 115M] freq = 9832

and/or	472	0.7	adjective-of	4865	4.6
بها	174	12.6	ضوء	456	10.48
أصفر	52	9.52	شاي	178	10.14
برتقالي	9	8.45	لون	300	9.42
أزرق	13	8.31	خط	355	8.94
أحمر	77	7.87	سطح	61	8.54
أبيض	42	6.99	منطقة	637	8.41
أسود	12	6.36	جبل	119	8.35
			حرارة	57	8.23
			عشب	51	8.1
			مسيرة	81	7.76
			صباح	67	7.65
			نبات	38	7.25

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Chinese GigaWord 2 Corpus: Mainland, simplified freq = **6196** (24.8 per million)

SendObject of	134	25
结束	18	6.41
停止	26	4.77
专门	10	3.01
继续	14	1.38
开始	11	1.21

Concordance – examples of a word, lemma, phrase in context



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doc#486850	from the airport centre. Why can't dubson get caught with his pants down, or how about falwell
doc#1398336	similar." Looking like a man who has been caught with his pants down and is valiantly trying
doc#4315657	anyhow, and if you're stupid enough to get caught with your digital pants down, who is you
doc#5684214	without a court order. But I guess he wasn't caught with his pants down in the oval office...
doc#9260036	the fuck is going on?" he says jealously. Caught with his pants down, Sergio throws Camille
doc#11238834	wright Former Professional not long ago was caught with his pants down and destroyed a lovely
doc#11813476	rescue you and will probably think we've been caught with our pants down. There going to come
doc#11986224	rights because few had been engaged "we were caught with our pants down," he said. "September
doc#12513909	, I'll have to say I was quite literally caught with my pants down. The saga that began
doc#12573174	, or simply knew about it, or was simply caught with its pants down, we may never know.
doc#14864966	Thanks Donna Ah well now.... another one caught with his pants down. But wait I...could
doc#16638028	you routinely make fun of people who get caught with their pants down. I think all of you
doc#16662568	appears that its the Liberals who have been caught with thier pants down in regards to the

tea ^(noun)
enTenTen12 freq = 614,038 (47.3 per million)

Parallel corpus – bilingual corpus, for translation examples



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OPUS2 Chinese Simplified

我想我能叫 Lee Marvin 還有 Van Cleef 的
微笑

好了，到了吧，讓我們 微笑 的...

我希望能在普通民? 众 的? 臉上看到 微笑

我相信她一定在上面向我們? 們 微笑
如果我 微笑? 着

她躺在我胸口 微笑 的樣子

“ 向上 微笑 ”

?? 錯，我? 還是值得? 別人? 對我 微笑 并
且? 對我? 說我 很美?? 动人的

愿 神? 对你 微笑，再? 见了

并且 永?? 记住 微笑 比子? 彈 便宜

OPUS2 English

I guess I could be Lee Marvin with a Van Cleef **smile** .

" Let's just sweep everything under the rug with a pat on the back, a **smile** and don't forget the glass of champagne to take the edge off. "

I hope the average Joe will be able to keep the **smile** on his face.

Lily: I'm sure she's **smiling** down on us right now.

What if I were **smiling** and running into your arms?

My Maria and ... her little **smile** when she laid on my chest.

And why is no one ever says, " I think he's down there now, **smiling** up at us. "

Yeah, I deserve to have somebody **smile** at me and tell me I'm pretty.

Well, may the gods **smile** on you. See you around.

And always remember that a **smile** is cheaper than a bullet.

WebBootCat – create corpus from web-pages



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WebBootCaT: Create corpus

[Get seed words from Wikipedia](#)

Corpus name

Language English ▾

Creating BootCaT corpora is available only for those language which we can at least tokenise. All such languages are listed here.

Build word sketches ☒

This option only has effect if a pre-loaded sketch grammar is available for the selected language.

Input type

☒ Seed words
☐ URLs

Select "URLs" to download data from specified URLs rather than use seed words for finding the URLs.

Seed words

volcanic eruption lava cone magma volcano
ash vent flow pyroclastic explosive
crater tephra caldera plume Stromboli

Random tuples will be selected from the seed words to query a search engine. Input 3 to 20 words or multi-word expressions. Use space as separator. Enclose multi-word expressions into quotes ("").

Compile corpus when ☒

finished

Automatically compile corpus when WebBootCaT processing is finished.

[Show advanced options](#)

Terminology – key words and multi-word terms in a specialism



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Vulcanology_en: Extracted keywords and terms

Keywords

- lava (62.3, 3249)
- eruption (55.6, 2939)
- volcano (45.9, 2463)
- ash (42.5, 2333)
- eruptions (41.2, 2167)
- volcanic (36.7, 1979)
- magma (36.0, 1870)
- vent (29.3, 1547)
- crater (25.5, 1385)
- flows (24.9, 1449)
- volcanoes (23.3, 1208)
- flow (21.7, 1632)
- pyroclastic (20.4, 1029)
- tephra (20.4, 1026)
- caldera (20.2, 1027)
- cone (19.3, 1005)
- plume (17.8, 906)
- explosive (17.6, 949)
- vents (12.5, 612)
- cinder (12.3, 601)
- strombolian (11.8, 573)
- hawaiian (5.9, 270)
- kilauea (5.9, 261)
- kilometers (5.9, 287)
- active (5.8, 492)
- miles (5.7, 488)
- basaltic (5.7, 248)
- molten (5.5, 245)
- mantle (5.5, 251)
- particles (5.5, 283)
- volume (5.2, 375)
- lavas (5.2, 222)
- steam (5.1, 252)
- seismic (5.1, 228)
- viscosity (5.1, 221)
- occurred (5.1, 344)
- crust (5.0, 223)
- explosion (4.9, 240)
- ejected (4.9, 209)
- earthquakes (4.8, 214)
- zone (4.8, 285)
- tremor (4.8, 204)
- shield (4.8, 229)

Terms

- lava flow (9.9)
- cinder cone (6.2)
- pyroclastic flow (5.7)
- volcanic ash (5.4)
- eruption column (4.3)
- magma chamber (4.3)
- explosive eruption (4.1)
- volcanic eruption (4.0)
- ash plume (4.0)
- differential flow (3.9)
- lava lake (3.8)
- lava dome (3.1)
- volcanic activity (3.1)
- rift zone (3.1)
- strombolian activity (3.0)
- east rift (2.9)
- molten rock (2.7)
- pyroclastic material (2.7)
- se crater (2.6)
- explosive activity (2.6)
- east rift zone (2.5)
- flow vent (2.5)

More SketchEngine functions



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Word lists – frequency lists filtered with user criteria

n-grams – frequency lists of multi-word expressions

Collocations – word co-occurrence analysis

Word sketch difference – compares two words by collocations

Diachronic analysis (Trends) – word usage change over time

Part-of-Speech tagging and lemmatization, in many languages



- Use SkE to collect and analyse text data from the Web
- Leeds Internet Corpora with seed-words for a large corpus
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Let's try them ...



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- SketchEngine www.sketchengine.eu
- Leeds Internet Corpora corpus.leeds.ac.uk/internet.html
- ACL SIGWAC www.sigwac.org.uk

SketchEngine licence is for on-campus use, or via VPN

it.leeds.ac.uk How to connect to the University VPN:

it.leeds.ac.uk/it?id=kb_article&sysparm_article=KB0014410

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